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Spatially Adaptive Multi-Frequency Graph Representation and Evolution for Hyperspectral Image Clustering

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***Abstract*—Hyperspectral image clustering aims to discover land-cover category structures in an unsupervised manner. Graph-based methods have shown strong potential for this task by modeling spatial topological relationships and non-Euclidean correlations among samples. However, existing graph-based clustering methods often rely on smooth propagation or introduce frequency responses through graph operators and globally shared weights, which may be insufficient to capture the spatially varying frequency demands of hyperspectral scenes. In particular, homogeneous regions usually require low-frequency consistency enhancement, whereas boundaries, mixed pixels, and fine-grained structures demand stronger high-frequency discrimination. To address this issue, we propose a spatially adaptive multi-frequency graph representation and evolution framework (SAMFGE) for hyperspectral image clustering. Specifically, we design a collaborative multi-frequency representation module to capture low-frequency homogeneity and high-frequency heterogeneity. A node- and channel-aware frequency coordination mechanism is then introduced to adaptively assign frequency preferences according to local structural properties and discriminative spectral-spatial responses. Furthermore, the learned multi-frequency cues are exploited as interpretable evidence to guide graph evolution, enabling representation learning and topology refinement to co-evolve in a frequency-aware manner. Experimental results on multiple public hyperspectral datasets demonstrate the effectiveness of the proposed framework.**

***Index Terms*—Hyperspectral image clustering, graph representation learning, graph signal processing, multi-frequency graph filtering, adaptive frequency coordination, graph topology evolution.**

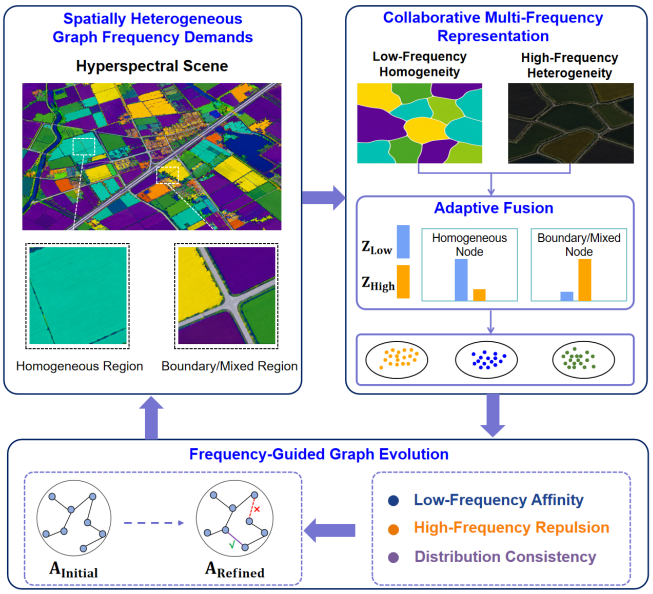


Fig. 1. Conceptual illustration of the proposed spatially adaptive multi-frequency graph representation and evolution framework. Hyperspectral scenes contain spatially heterogeneous regions with different graph-frequency demands, making it challenging for fixed or globally shared graph propagation to simultaneously preserve homogeneous consistency and boundary-sensitive discrimination. The proposed method addresses this dilemma by adaptively coordinating low-frequency consistency and high-frequency discrimination, and further exploits the learned multi-frequency cues to guide graph topology refinement.

I. Introduction

Hyperspectral images record the reflectance properties of ground objects across hundreds of contiguous narrow spectral bands, thereby providing abundant spectral features and rich spatial contextual information [1], [2], [3]. Owing to these advantages, they have shown great potential in a wide range of applications, including agricultural monitoring [4], environmental sensing [5], mineral exploration [6], and urban analysis [7]. Unlike supervised classification methods that rely on pixel-level annotations [8], [9], hyperspectral image clustering aims to uncover the latent land-cover category structure without labeled data, making it particularly valuable in remote sensing scenarios where annotation is expensive and prior knowledge is limited. However, hyperspectral data are typically characterized by high-dimensional redundancy, noise sensitivity, and complex spatial distributions, which makes learning stable and discriminative spectral–spatial representations under unsupervised settings a fundamental challenge in this field.

To address the above challenges, hyperspectral image clustering methods have evolved from traditional shallow models to deep representation learning. Among different technical routes, graph-based modeling has become an important direction, because it explicitly captures the spatial topological

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relationships and non-Euclidean correlations among samples, and is therefore well suited to describing the complex spectral–spatial structure of hyperspectral scenes. In particular, graph construction schemes based on superpixel graphs [10], [11], anchor graphs [12], [13], bipartite graphs [14], and hypergraphs [15] have further alleviated the computational burden of pixel-level graph learning, making it possible to achieve a better balance between structural expressiveness and computational scalability.

Although graph-based methods have achieved promising progress, many existing graph propagation mechanisms still rely heavily on local consistency preservation or smooth information aggregation. This design is effective for enhancing homogeneous regions, but it implicitly assumes that different spatial structures can be processed with similar propagation preferences. Such an assumption is not always suitable for hyperspectral scenes. In real hyperspectral images, homogeneous regions, category boundaries, transition areas, mixed pixels, and fine-grained structures usually exhibit different spatial patterns and therefore require different graph-frequency responses. If a globally shared or insufficiently adaptive propagation mechanism is applied to all regions, homogeneous areas may benefit from smoothing, but boundary-sensitive and structurally heterogeneous cues may be weakened. Therefore, the key limitation of existing graph propagation is not merely oversmoothing, but the insufficient adaptation of graph-frequency preference to spatially heterogeneous hyperspectral structures.

From the graph-frequency perspective, homogeneous regions usually correspond to slowly varying graph signals and benefit from low-frequency aggregation, which can enhance intra-class consistency and suppress local noise. In contrast, category boundaries, transition regions, mixed pixels, and fine-grained land-cover structures are often associated with abrupt local variations and require stronger high-frequency discrimination to preserve structural differences. This observation suggests that hyperspectral image clustering should not depend only on a single smooth propagation process. Instead, it requires a spatially adaptive multi-frequency coordination mechanism that can jointly exploit low-frequency homogeneity and high-frequency heterogeneity according to local structural properties. Moreover, different spectral-spatial channels may contribute unequally to clustering discrimination, which further indicates that frequency preferences should be coordinated not only at the node level but also at the channel level.

Recent studies [16] have introduced low-frequency and high-frequency graph modeling into hyperspectral clustering, demonstrating the usefulness of multi-frequency information for improving clustering representations. Nevertheless, existing frequency-aware methods often control frequency responses at the graph operator level or through globally shared weights. Such designs provide useful frequency modeling ability, but they are still insufficient to reflect the fact that different graph nodes and different spectral-spatial channels may require different frequency emphasis. As a result, homogeneous regions, boundary regions, and mixed regions may still be processed with insufficiently differentiated frequency responses, which limits the model's ability to simultaneously preserve regional consistency and structural discriminability. In addition, multi-frequency cues are usually used mainly for representation learning, while their potential value for graph topology refinement has not been fully explored. Since low-frequency affinity can indicate reliable homogeneous connections and high-frequency discrepancy can reveal potential structural transitions, multi-frequency information can also provide interpretable evidence for refining unreliable graph edges.

Motivated by the above analysis, we propose a spatially adaptive multi-frequency collaborative clustering framework for hyperspectral images. The proposed method represents each hyperspectral image as a region-level graph and learns complementary frequency-aware representations to capture both homogeneous consistency and heterogeneous structural cues. To avoid imposing a fixed frequency preference on all regions, a node- and channel-aware frequency coordination mechanism is introduced to adaptively integrate low-frequency and high-frequency information according to local structural properties and discriminative spectral-spatial responses. Moreover, the learned multi-frequency cues are further used to guide graph evolution, so that representation learning and topology refinement can be jointly optimized in a frequency-aware manner. Extensive experiments on multiple public hyperspectral datasets demonstrate the effectiveness of the proposed method. Quantitative comparisons, visual clustering maps, ablation studies, and parameter analyses verify the advantages of adaptive frequency coordination and frequency-guided graph evolution.

The main contributions of this paper are summarized as follows:

- A unified spatially adaptive multi-frequency graph representation and evolution framework. We propose a spatially adaptive multi-frequency graph representation and evolution framework for hyperspectral image clustering. By formulating clustering on a region-level graph, the framework integrates complementary multi-frequency representation learning, adaptive frequency coordination, and frequency-guided topology evolution into a unified learning process. Compared with existing graph propagation schemes that often rely on fixed or globally shared frequency preferences, the proposed framework provides a more flexible way to model spatially heterogeneous hyperspectral structures.
- A node- and channel-aware frequency coordination mechanism. We design a collaborative multi-frequency representation module to capture complementary low-frequency and high-frequency information. A node- and channel-aware coordination mechanism is further introduced to adaptively assign frequency preferences according to local structural properties and discriminative spectral-spatial responses. This enables the model to better balance regional consistency and structural discriminability in different hyperspectral regions.
- Frequency-guided graph evolution strategy. We introduce a frequency-guided graph evolution strategy that reuses the learned multi-frequency cues as interpretable structural evidence for edge updating. Low-frequency affinity,

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high-frequency discrepancy, and clustering distribution consistency are jointly used to regularize topology refinement, which improves graph propagation reliability and enhances clustering performance.

II. RELATED WORK

Hyperspectral image clustering aims to discover latent land-cover category structures from unlabeled data, and has been widely studied from traditional shallow models, deep representation learning, and graph-based modeling perspectives. Traditional methods [17], [18], [19] mainly rely on raw spectral features, self-expression coefficients, or shallow similarity measures, while deep methods further improve clustering quality by learning more discriminative latent representations [20], [21], [22], [23]. Among these lines, graph-based modeling has become one of the most important directions because it can explicitly characterize spatial topological relationships and non-Euclidean correlations among samples, which is particularly suitable for complex hyperspectral scenes. Since the proposed method is built upon graph-based clustering and further focuses on multi-frequency representation modeling and graph structure optimization, we briefly review the most relevant studies from these three aspects.

A. Graph-Based Hyperspectral Image Clustering

Graph-based hyperspectral image clustering [24], [25], [26], [27] has become a particularly important research direction in recent years. Compared with methods that partition samples solely based on raw spectral features or shallow similarities, graph models provide a more natural way to characterize complex spectral–spatial structures by explicitly encoding sample relationships. To improve scalability, existing studies have introduced superpixel graphs [28], anchor graphs [29], bipartite graphs [30], and hypergraphs [31] to reduce the computational burden of pixel-level graph learning while maintaining structural expressiveness. These methods have substantially improved clustering performance. Nevertheless, their core representation learning mechanisms are still largely driven by local consistency preservation and smooth propagation, which makes them more effective at modeling homogeneous regions than at preserving structurally heterogeneous cues.

Recent studies in heterophily-aware graph learning provide a complementary perspective by preserving informative differences between neighboring nodes through higher-order neighborhood aggregation, adaptive message propagation, and differentiated neighborhood modeling [32], [33], [34]. From a graph-frequency perspective, these local differences can be associated with higher-frequency components that complement smooth regional information. However, the adaptation of heterophily-aware learning to the spectral–spatial characteristics and unsupervised clustering setting of hyperspectral images remains relatively limited.

B. Multi-Frequency Information Modeling for Hyperspectral Clustering

To better account for the coexistence of regional homogeneity and structural heterogeneity in hyperspectral scenes, recent studies [16], [35] have introduced graph-frequency modeling into hyperspectral clustering. Low-frequency components are useful for suppressing noise and preserving regional consistency, whereas high-frequency components help retain boundary-sensitive and locally heterogeneous structures. These studies verify the importance of multi-frequency information for improving clustering representations. Beyond predefined graph-frequency decomposition, recent studies on adaptive frequency learning have further shown that the contributions of different frequency components can be adjusted based on graph structure or node characteristics through learnable propagation coefficients, gating mechanisms, or homophily-aware graph filtering [36], [37]. These studies suggest that graph representation learning can benefit from both complementary frequency information and adaptive coordination of its contributions across diverse structural patterns. However, the key issue is not only whether low and high-frequency information is used, but whether their relative importance can be adaptively determined according to local spatial structures and feature channels. Existing hyperspectral clustering methods usually define frequency responses through specific graph operators or balance different frequency components with shared parameters, which provides useful frequency awareness but may not sufficiently reflect the spatially varying frequency demands of different hyperspectral regions. Therefore, more adaptive multi-frequency coordination remains necessary for simultaneously preserving regional consistency and structural discriminability.

C. Graph Structure Optimization and Edge Learning

Besides representation learning, graph reliability itself also has a significant impact on hyperspectral clustering results. Since hyperspectral graphs are usually constructed from spatial adjacency, spectral similarity, or their variants, erroneous cross-category connections are often unavoidable. Once such unreliable edges are repeatedly involved in graph propagation, they may cause cross-boundary information leakage, feature mixing, and reduced class separability. To alleviate this issue, existing studies [16], [38] have attempted to refine the initial graph through edge learning, evidence-guided refinement, or task-driven updating. These efforts show that graph topology optimization is beneficial for improving clustering quality [39], [40]. Nevertheless, current graph updating mechanisms are still mainly driven by current embeddings, similarity estimation, or clustering states. Such criteria are useful but may not explicitly reflect the local frequency characteristics of hyperspectral structures. Since low-frequency similarity indicates homogeneous consistency and high-frequency responses often reveal structural transitions, multi-frequency cues can provide complementary and interpretable evidence for graph refinement.

D. Position of This Work

The above studies show that both multi-frequency modeling and graph refinement are useful for hyperspectral clustering. However, existing methods have not fully addressed how to

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adapt frequency preferences to spatially heterogeneous hyperspectral structures. In particular, homogeneous regions, boundary regions, mixed regions, and fine-grained structures may require different balances between low-frequency consistency and high-frequency discrimination, rather than being processed with fixed or globally shared frequency responses. Different from existing studies, this work focuses on spatially adaptive multi-frequency coordination. Specifically, we learn complementary representations that characterize low-frequency homogeneous consistency and high-frequency structural heterogeneity, and dynamically integrate them according to node-level structural properties and channel-level discriminative responses. The learned frequency cues are further reused as interpretable auxiliary evidence for graph edge evolution. In this way, the proposed method centers on adaptive multi-frequency representation learning, while incorporating frequency-guided graph refinement as a complementary mechanism to improve propagation reliability.

III. PROPOSED METHOD

This section presents the proposed spatially adaptive multi-frequency graph representation and evolution framework for hyperspectral image clustering. The framework is designed to address the spatially heterogeneous frequency demands of hyperspectral scenes, where homogeneous regions require low-frequency consistency, while boundaries, transition regions, mixed pixels, and fine-grained structures require high-frequency discrimination. To this end, the proposed method constructs a compact region-level graph, learns complementary multi-frequency representations, adaptively coordinates them at the node and channel levels, and further exploits the learned frequency cues to guide graph topology evolution. The overall framework consists of four components: region-level graph construction, collaborative multi-frequency representation learning, node- and channel-aware frequency coordination, and frequency-guided graph evolution.

A. Graph Construction and Preprocessing

Considering that pixel-level graph construction in hyperspectral images usually suffers from high computational complexity and is also vulnerable to local noise interference, this paper adopts a region-level graph modeling paradigm to preprocess the original image. Given an input hyperspectral image $\mathbf{X}_{\mathrm{raw}} \in \mathbb{R}^{H\times W\times B}$, principal component analysis (PCA) is first applied to reduce the spectral dimensionality, yielding a reduced representation $\mathbf{X}_{\mathrm{pca}} \in \mathbb{R}^{H\times W\times d}$. Subsequently, the simple linear iterative clustering (SLIC) [41] superpixel segmentation algorithm is performed on the reduced image to partition it into N locally homogeneous regions $\{S_1, S_2, \ldots, S_N\}$. Based on the above region partition, an undirected graph $\mathcal{G} = (\mathcal{V}, \mathbf{A})$ is constructed, where each superpixel corresponds to a graph node. The node feature matrix is represented by the mean feature of all pixels within the corresponding region:

$$(\mathbf{X}_{\mathrm{s}})_i = \frac{1}{|S_i|}\sum_{p\in S_i}\mathbf{X}_{\mathrm{pca}}(p). \tag{1}$$

TABLE I
SUMMARY OF MAIN NOTATIONS

Notation	Meaning
$\mathbf{X}_{\mathrm{raw}} \in \mathbb{R}^{H\times W\times B}$	Raw hyperspectral image
$\mathbf{X}_{\mathrm{pca}} \in \mathbb{R}^{H\times W\times d}$	PCA-reduced hyperspectral image
$G = (V, A)$	Region-level graph with node set V and adjacency matrix A
N	Number of superpixel nodes
$\mathbf{X}_s \in \mathbb{R}^{N\times d}$	Superpixel-level node feature matrix
$\mathbf{A}^{(0)}, \mathbf{A}^{(t)} \in \mathbb{R}^{N\times N}$	Initial and evolving adjacency matrices
$\mathbf{Q} \in \{0,1\}^{HW\times N}$	Pixel-to-superpixel assignment matrix
$\mathbf{L}_{\mathrm{sym}}, \mathbf{L}_{\mathrm{rw}}$	Symmetric normalized and random-walk graph Laplacians
K	Chebyshev polynomial order
$\mathbf{Z}_{\mathrm{low}}, \mathbf{Z}_{\mathrm{high}} \in \mathbb{R}^{N\times d}$	Low- and high-frequency node representations
$\mathbf{Z}_{\mathrm{fused}} \in \mathbb{R}^{N\times d}$	Fused multi-frequency node representation
$\mathbf{Q}_{\mathrm{prob}} \in \mathbb{R}^{N\times C}$	Node-wise clustering probability matrix
C	Number of clusters
$\mathbf{W}_{\mathrm{pre}}, \mathbf{W}_{\mathrm{emp}} \in \mathbb{R}^{N\times N}$	Predicted and empirical edge-weight matrices
$\mathbf{W}_{\mathrm{edge}}$	Learnable relation metric for edge prediction
$\mathbf{M} \in \{0,1\}^{N\times N}$	Initial local-connectivity mask matrix
$\hat{\mathbf{Y}} \in \{1,\ldots,C\}^N$	Fixed pseudo-label vector

The initial adjacency matrix $\mathbf{A}^{(0)} \in \mathbb{R}^{N\times N}$ is defined according to the spatial adjacency relationships between superpixels. Specifically, two nodes are connected if their corresponding superpixels share a common boundary:

$$\mathbf{A}^{(0)}_{i,j} = \begin{cases} 1, & \text{if } S_i \text{ is adjacent to } S_j, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

In addition, an assignment matrix $\mathbf{Q} \in \{0,1\}^{HW\times N}$ is defined to record the affiliation relationship between pixels and superpixels, which is used to map the subsequent region-level clustering results back to the pixel space. After preprocessing, the original hyperspectral image is represented as a region-level graph. This graph reduces the computational burden of pixel-level modeling and suppresses local pixel-wise noise by treating each superpixel as a node. It also provides the structural basis for subsequent multi-frequency representation learning and graph topology evolution.

B. Collaborative Multi-Frequency Representation Learning

Different hyperspectral structures require different graph-frequency responses. Homogeneous regions benefit from smooth low-frequency aggregation, whereas boundaries, transition regions, mixed pixels, and fine-grained structures require stronger responses to local variations. Therefore, a single smooth propagation process is insufficient for simultaneously preserving regional consistency and structural discriminability. We thus design a collaborative multi-frequency representation learning module to extract complementary frequency-aware cues from the region-level graph.

From the perspective of graph filtering, graph representation learning can be viewed as a frequency-selective transformation of graph signals [42]. Given a graph operator $\mathbf{L}$ and an input signal $\mathbf{x}$, a graph filter can be generally expressed as

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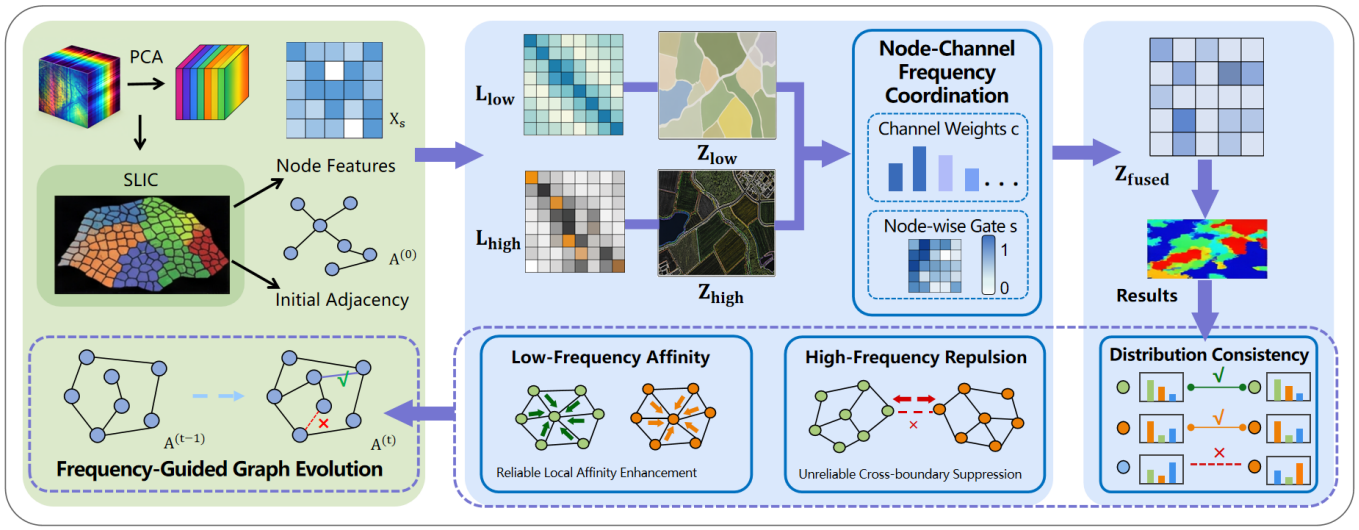


Fig. 2. Overall architecture of the proposed framework (SAMFGE). The method constructs a region-level graph, learns collaborative multi-frequency representations, performs node- and channel-aware frequency coordination, and uses the learned frequency cues to guide graph topology evolution for hyperspectral image clustering.

$g(\mathbf{L})\mathbf{x}$, where $g(\cdot)$ denotes the spectral response. To avoid the high computational cost of explicit eigendecomposition while retaining learnable localized filtering capability, we employ Chebyshev polynomial approximation [43], [44] to parameterize the graph filter:

$$g_\theta(\mathbf{L})\mathbf{x} \approx \sum_{k=0}^{K} \theta_k \, T_k(\tilde{\mathbf{L}})\mathbf{x}, \tag{3}$$

where $\tilde{\mathbf{L}}$ is the rescaled graph operator, $T_k(\cdot)$ denotes the k-th order Chebyshev polynomial, and θ_k is the learnable coefficient. This formulation provides a localized K-hop receptive field and enables learnable frequency selective transformations without explicit eigendecomposition. Based on it, we construct a collaborative multi-frequency representation module, where one branch focuses on low-frequency smooth consistency and the other branch enhances high-frequency heterogeneous responses. The resulting complementary representations are then dynamically integrated through an adaptive frequency coordination fusion mechanism.

1) Low-Frequency Homogeneity Modeling: Most existing graph propagation schemes strengthen intra-region consistency by encouraging smoothness over neighboring nodes, which essentially corresponds to preserving low-frequency graph components. This is particularly useful for modeling homogeneous regions in hyperspectral images, where stable local aggregation helps suppress noise and maintain regional coherence. To explicitly preserve such low-frequency information, we construct a low-frequency branch to model smooth consistency within regions.

Specifically, this branch employs the symmetrically normalized Laplacian operator

$$\mathbf{L}_{\mathrm{sym}} = \mathbf{I} - \widetilde{\mathbf{D}}^{-\frac{1}{2}} \widetilde{\mathbf{A}} \widetilde{\mathbf{D}}^{-\frac{1}{2}}, \tag{4}$$

where $\widetilde{\mathbf{A}} = \mathbf{A}+\mathbf{I}$ denotes the augmented adjacency matrix and $\widetilde{\mathbf{D}}$ is the corresponding degree matrix. Compared with other normalization forms, $\mathbf{L}_{\mathrm{sym}}$ yields more stable and balanced neighborhood aggregation, and is therefore more suitable for characterizing low-frequency smooth structures with relatively mild intra-region variations. Following the implementation setting, the corresponding rescaled operator is written as

$$\tilde{\mathbf{L}}_{\mathrm{sym}} = \mathbf{L}_{\mathrm{sym}} - \mathbf{I}. \tag{5}$$

To encourage a low-frequency response at the early stage of training, a low-pass spectral response initialization prior is imposed on the learnable Chebyshev coefficients $\boldsymbol{\theta}_{\mathrm{low}} \in \mathbb{R}^{K+1}$. Under this design, the low-frequency smooth representation $\mathbf{Z}_{\mathrm{low}} \in \mathbb{R}^{N\times d}$ is computed as

$$\mathbf{Z}_{\mathrm{low}} = \sum_{k=0}^{K} \theta_{\mathrm{low},k} \, T_k\left(\tilde{\mathbf{L}}_{\mathrm{sym}}\right) \mathbf{X}_{\mathrm{s}}. \tag{6}$$

This branch captures smooth intra-region consistency and provides low-frequency affinity cues for both adaptive frequency coordination and reliable edge preservation during graph evolution.

2) High-Frequency Heterogeneity Modeling: Although low-frequency propagation preserves homogeneous consistency, it may attenuate local variations around boundaries, transition regions, mixed pixels, and fine-grained structures. To compensate for these weakened cues, we construct a high-frequency branch to enhance local structural responses on the graph.

Specifically, this branch adopts the random-walk Laplacian operator

$$\mathbf{L}_{\mathrm{rw}} = \mathbf{I} - \widetilde{\mathbf{D}}^{-1}\widetilde{\mathbf{A}}. \tag{7}$$

Compared with $\mathbf{L}_{\mathrm{sym}}$, $\mathbf{L}_{\mathrm{rw}}$ exhibits more flexible response characteristics to local topological changes during propagation, the node-wise operation measures the deviation of each node from its degree-normalized neighborhood aggregation. Therefore, locally homogeneous regions tend to exhibit relatively small deviations, whereas boundaries, transition regions, and mixed regions generally produce stronger local discrepancies.

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This variation-sensitive property provides a natural graph-spectral basis for modeling structural heterogeneity [16], [36], [45]. The corresponding rescaled operator is defined as

$$\tilde{\mathbf{L}}_{\mathrm{rw}} = \mathbf{L}_{\mathrm{rw}} - \mathbf{I}. \tag{8}$$

Rather than directly learning an unconstrained high-frequency filter, we construct a differential spectral response [46] using two sets of Chebyshev coefficients [43], namely $\boldsymbol{\theta}_{\mathrm{wide}}$ and $\boldsymbol{\theta}_{\mathrm{narrow}}$. To establish distinct initial response behaviors, $\boldsymbol{\theta}_{\mathrm{wide}}$ and $\boldsymbol{\theta}_{\mathrm{narrow}}$ are asymmetrically initialized from different coefficient ranges. A freely learned high-frequency filter may overlap with the low-frequency branch and repeatedly capture smooth components. In contrast, subtracting the narrow response from the wide response suppresses their shared components while retaining differential components associated with local feature variations. This provides a mild inductive bias for learning boundary-sensitive and locally heterogeneous cues, making the high-frequency branch more complementary to the low-frequency branch. Accordingly, the output of the high-frequency branch, denoted by $\mathbf{Z}_{\mathrm{high}} \in \mathbb{R}^{N\times d}$, is given by

$$\mathbf{Z}_{\mathrm{high}} = \sum_{k=0}^{K} \left(\theta_{\mathrm{wide},k} - \theta_{\mathrm{narrow},k}\right) T_k\left(\tilde{\mathbf{L}}_{\mathrm{rw}}\right) \mathbf{X}_{\mathrm{s}}. \tag{9}$$

The resulting representation provides high-frequency discrepancy cues, complementing low-frequency affinity in preserving local details and suppressing unreliable cross-boundary propagation.

3) Node- and Channel-Aware Frequency Coordination: Given the complementary low-frequency and high-frequency representations, the key issue is to determine suitable frequency preferences for different nodes and feature dimensions. A fixed or globally shared balance between the two branches is insufficient for spatially heterogeneous hyperspectral scenes, since homogeneous regions, boundary regions, and mixed regions often require different frequency emphasis. Meanwhile, different spectral-spatial channels may contribute unequally to clustering. Therefore, we coordinate the two frequency branches at both the channel and node levels.

We first perform channel-wise calibration to emphasize discriminative spectral-spatial response dimensions. $\mathbf{Z}_{\mathrm{low}}, \mathbf{Z}_{\mathrm{high}} \in \mathbb{R}^{N\times d}$ denote the low- and high-frequency representations. The two frequency representations are first summed and then averaged over the node dimension to obtain a global channel descriptor $\mathbf{g} \in \mathbb{R}^d$:

$$\mathbf{g} = \frac{1}{N}\sum_{i=1}^{N} \left(\mathbf{z}_{\mathrm{low},i} + \mathbf{z}_{\mathrm{high},i}\right), \tag{10}$$

$\mathbf{g}$ is then used to estimate channel importance $\mathbf{c} \in \mathbb{R}^d$:

$$\mathbf{c} = \sigma\left(\mathrm{MLP}_c(\mathbf{g})\right), \tag{11}$$

The channel calibrated features are then obtained as

$$\mathbf{Z}_{\mathrm{low}}^{(c)} = \mathbf{Z}_{\mathrm{low}} \odot \left(\mathbf{1}_N \mathbf{c}^\top\right), \mathbf{Z}_{\mathrm{high}}^{(c)} = \mathbf{Z}_{\mathrm{high}} \odot \left(\mathbf{1}_N \mathbf{c}^\top\right). \tag{12}$$

After channel level recalibration, different graph nodes still require different multi-frequency preferences due to their distinct structural properties. On this basis, node-level adaptive gating is further introduced along the spatial dimension to dynamically assign the importance of the two branches according to the local feature states of different nodes. Specifically, the calibrated dual-branch features are concatenated along the channel dimension, and a spatial gating network is employed to generate node-wise weights $\mathbf{s} \in \mathbb{R}^{N\times 1}$:

$$\mathbf{s} = \sigma\Big(\mathrm{MLP}_{\mathrm{s}}\Big(\Big[\mathbf{Z}_{\mathrm{low}}^{(\mathrm{c})} \,\|\, \mathbf{Z}_{\mathrm{high}}^{(\mathrm{c})}\Big]\Big)\Big), \tag{13}$$

where $\mathbf{s}$ characterizes the degree to which each node relies on low-frequency smooth information in the final fused representation, while $\mathbf{1}-\mathbf{s}$ corresponds to the relative contribution of high-frequency structural information. Accordingly, the final fused representation $\mathbf{Z}_{\mathrm{fused}} \in \mathbb{R}^{N\times d}$ is defined as

$$\mathbf{Z}_{\mathrm{fused}} = \left(\mathbf{s}\mathbf{1}_d^\top\right) \odot \mathbf{Z}_{\mathrm{low}}^{(c)} + \left[(\mathbf{1}_N - \mathbf{s})\mathbf{1}_d^\top\right] \odot \mathbf{Z}_{\mathrm{high}}^{(c)}. \tag{14}$$

Through this coordination process, the model adaptively balances low-frequency consistency and high-frequency structural information across different regions and response dimensions. The fused representation improves clustering discriminability and serves as the basis for frequency-guided graph evolution.

C. Frequency-Guided Graph Evolution

Although the region-level graph provides a compact topology, the initial graph constructed from superpixel adjacency may still contain unreliable cross-category connections, especially around boundaries and mixed regions. Such edges may cause cross-boundary information leakage during propagation. The fused multi-frequency representation provides useful structural evidence for graph refinement: low-frequency cues indicate homogeneous affinity, while high-frequency responses reveal potential boundary or transition structures. Based on this observation, we introduce a frequency-guided graph evolution module to refine graph topology using the learned frequency cues.

This module follows a prediction-guidance design. A learnable edge prediction matrix $\mathbf{W}_{\mathrm{pre}}$ is estimated from the fused representation to keep topology updating adaptive to the current node embeddings. Meanwhile, an empirically guided matrix $\mathbf{W}_{\mathrm{emp}}$ is constructed from low-frequency affinity, high-frequency boundary penalty, and clustering distribution consistency to provide interpretable structural regularization. During training, $\mathbf{W}_{\mathrm{emp}}$ regularizes $\mathbf{W}_{\mathrm{pre}}$, while $\mathbf{W}_{\mathrm{pre}}$ is used for graph updating. This design makes graph evolution both representation-adaptive and structurally interpretable.

To avoid imposing a fixed similarity assumption through direct inner products, we introduce a learnable relation metric $\mathbf{W}_{\mathrm{edge}}$ to estimate pairwise connectivity tendencies. The predicted edge-weight matrix is formulated as follows:

$$\mathbf{W}_{\mathrm{pre}} = \sigma\left(\mathbf{Z}_{\mathrm{fused}} \mathbf{W}_{\mathrm{edge}} \mathbf{Z}_{\mathrm{fused}}^\top\right), \tag{15}$$

which describes the data-driven connectivity tendency between nodes under the current multi-frequency representation.

Following the above prediction-guidance design, we next detail the construction of the empirically guided matrix $\mathbf{W}_{\mathrm{emp}}$. The reliability of a local edge is evaluated from three complementary aspects. First, two incident nodes are expected to

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exhibit consistent low-frequency responses if they belong to the same homogeneous region. Second, an edge should be treated cautiously when its incident nodes show strong high-frequency responses, since such responses are often associated with boundaries, transition regions, or mixed structures. Third, the two nodes should remain compatible in the clustering assignment space. Accordingly, we construct $\mathbf{W}_{\mathrm{emp}}$ by integrating three types of evidence, namely low-frequency affinity, high-frequency boundary penalty, and clustering distribution consistency.

Low-Frequency Affinity ($\mathbf{S}_{\mathrm{low}}$): The low-frequency branch captures smooth and consistent responses within homogeneous regions. Therefore, nodes with similar low-frequency features are more likely to belong to the same land-cover region and should maintain stronger connections. We measure this affinity by the cosine similarity of the low-frequency features $\mathbf{Z}_{\mathrm{low}}$:

$$\mathbf{S}_{\mathrm{low}} = \widetilde{\mathbf{Z}}_{\mathrm{low}} \widetilde{\mathbf{Z}}_{\mathrm{low}}^{\top}, \tag{16}$$

where $\widetilde{\mathbf{Z}}_{\mathrm{low}}$ is obtained by applying row-wise L_2 normalization to $\mathbf{Z}_{\mathrm{low}}$. This term encourages the preservation of reliable intra-region edges.

High-Frequency Repulsion ($\mathbf{P}_{\mathrm{high}}$): Nodes around boundaries, transition regions, and locally heterogeneous structures usually exhibit stronger high-frequency responses. Edges incident to such nodes are more likely to involve uncertain or cross-boundary propagation and should be treated cautiously. We therefore define a boundary penalty based on local high-frequency energy:

$$\mathbf{P}_{\mathrm{high},\,i,j} = \left(\|\mathbf{Z}_{\mathrm{high},\,i}\|_2 + \|\mathbf{Z}_{\mathrm{high},\,j}\|_2 \right). \tag{17}$$

This term characterizes structural discrepancy and serves to penalize unreliable edges that are likely to cross land-cover boundaries.

Distribution Consistency ($\mathbf{S}_{\mathrm{q}}$): Although low and high-frequency cues provide important structural evidence, edge evolution should also remain compatible with the current clustering semantics. To this end, we further measure the consistency between nodes in the clustering assignment space by computing the cosine similarity between their clustering probability distributions $\mathbf{Q}_{\mathrm{prob}}$:

$$\mathbf{S}_{\mathrm{q}} = \widetilde{\mathbf{Q}}_{\mathrm{prob}} \widetilde{\mathbf{Q}}_{\mathrm{prob}}^{\top}, \tag{18}$$

where $\widetilde{\mathbf{Q}}_{\mathrm{prob}}$ is obtained by applying row-wise L_2 normalization to $\mathbf{Q}_{\mathrm{prob}}$. This term provides complementary semantic-level guidance, encouraging graph evolution to remain compatible with class-level compactness.

By incorporating the three types of evidence, we construct the empirically guided matrix $\mathbf{W}_{\mathrm{emp}}$:

$$\mathbf{W}_{\mathrm{emp}} = \sigma\left(\frac{\mathbf{S}_{\mathrm{low}} - \mathbf{P}_{\mathrm{high}} + \mathbf{S}_{\mathrm{q}}}{\tau} \right) \odot \mathbf{M}, \tag{19}$$

where the binary mask matrix $\mathbf{M} \in \{0,1\}^{N \times N}$ represents the initial local connectivity and τ controls the sharpness of empirical guidance. This constraint preserves the sparse local topology prior of the initial graph and prevents the evolution process from deviating excessively from the original spatial adjacency structure. In essence, $\mathbf{W}_{\mathrm{emp}}$ provides an interpretable auxiliary criterion for deciding whether a local edge should be preserved, weakened, or relatively emphasized during graph evolution.

Algorithm 1 Training and Inference Procedure of the Proposed Framework.

Input: Raw HSI $\mathbf{X}_{\mathrm{raw}}$; superpixel scale parameter S; Chebyshev polynomial order K; learning rate η; training epochs T; graph update interval T_u; network $f(\cdot)$.
Output: Pixel-level clustering result $\mathbf{R}$.
Apply PCA and SLIC on $\mathbf{X}_{\mathrm{raw}}$ to obtain superpixel features $\mathbf{X}_{\mathrm{s}}$ and the initial adjacency matrix $\mathbf{A}^{(0)}$ and assignment matrix $\mathbf{Q}$.
Compute $\mathbf{L}_{\mathrm{sym}}$ and $\mathbf{L}_{\mathrm{rw}}$ from $\mathbf{A}^{(0)}$ by Eqs. (4) and (7), and generate pseudo-labels $\hat{\mathbf{Y}}$ by K-means on $\mathbf{X}_s$.
for $e = 1$ to T **do**
  Forward $f(\mathbf{X}_{\mathrm{s}}, \mathbf{L}_{\mathrm{sym}}, \mathbf{L}_{\mathrm{rw}})$ to obtain multi-frequency representations and clustering probabilities $\mathbf{Q}_{\mathrm{prob}}$.
  Compute $\mathbf{W}_{\mathrm{pre}}$ by Eq. (15) and construct $\mathbf{W}_{\mathrm{emp}}$ by Eq. (19), and update network parameters by minimizing $\mathcal{L}_{total}$ by Eq. (21).
  if $\mathrm{mod}(e, T_u) = 0$ **then**
    Update graph topology by Eq. (20) and recompute $\mathbf{L}_{sym}$ and $\mathbf{L}_{rw}$.
  end if
end for
Forward $f(\mathbf{X}_{\mathrm{s}}, \mathbf{L}_{\mathrm{sym}}, \mathbf{L}_{\mathrm{rw}})$ to obtain the final $\mathbf{Q}_{\mathrm{prob}}$.
Obtain the superpixel-level labels $\mathbf{R}_{\mathrm{sp}}$ by

$$\mathbf{R}_{\mathrm{sp}} = \arg\max(\mathbf{Q}_{\mathrm{prob}}, \mathrm{dim} = 1).$$

return the pixel-level clustering result $\mathbf{R}$, where each pixel is assigned the label of its associated superpixel according to $\mathbf{Q}$.

During training, the graph topology is updated according to the predicted edge weights:

$$\mathbf{A}^{(t+1)} = \gamma \mathbf{A}^{(t)} + (1-\gamma)\mathbf{W}_{\mathrm{pre}} \odot \mathbf{M}, \tag{20}$$

where $\gamma \in (0,1)$ is the momentum coefficient, the update follows a momentum-like form so that the evolved graph can gradually adapt to the learned representations rather than changing abruptly. Specifically, $\mathbf{W}_{\mathrm{pre}}$ is used for graph updating because it is a learnable edge prediction derived from the current representations, whereas $\mathbf{W}_{\mathrm{emp}}$ serves as interpretable guidance for regularization rather than a direct replacement of the graph topology. Meanwhile, the consistency between the learnable prediction $\mathbf{W}_{\mathrm{pre}}$ and the interpretable guidance $\mathbf{W}_{\mathrm{emp}}$ is enforced by the edge evolution loss defined in Section III-D. Subsequently, the graph filtering operators (e.g., $\mathbf{L}_{\mathrm{sym}}$ and $\mathbf{L}_{\mathrm{rw}}$) are dynamically reconstructed based on the updated adjacency matrix $\mathbf{A}^{(t+1)}$. In this way, the learned multi-frequency representations provide shared guidance for graph refinement, while the refined graph in turn reshapes subsequent frequency-aware propagation. Therefore, representation learning and graph topology evolution are not independently optimized, but progressively co-evolve during training.

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D. Joint Optimization Objective

To collaboratively optimize multi-frequency representation learning, clustering assignment refinement, and graph topology evolution within a unified framework, we define the overall objective as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pseudo}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{edge}}\mathcal{L}_{\text{edge}}, \tag{21}$$

Here, $\mathcal{L}_{\text{pseudo}}$ provides an initial clustering prior for unsupervised optimization, $\mathcal{L}_{\text{reg}}$ regularizes the validity and stability of clustering assignments, and $\mathcal{L}_{\text{edge}}$ aligns the learnable edge prediction with the interpretable multi-frequency guidance. The coefficients λ_{reg} and λ_{edge} balance clustering regularization and frequency-guided edge evolution.

First, k-means clustering is applied to the superpixel node features $\mathbf{X}_{\text{s}}$ to generate the pseudo-labels

$$\hat{\mathbf{Y}} \in \{1, 2, \ldots, C\}^N, \tag{22}$$

where C denotes the predefined number of clusters. The pseudo-labels generated from the initial node features are kept fixed during subsequent optimization. This fixed strategy provides a stable clustering reference for representation learning. Based on these pseudo-labels, a cross-entropy loss is adopted as a self-supervised signal to constrain the predicted clustering probability distribution $\mathbf{Q}_{\text{prob}} \in \mathbb{R}^{N\times C}$:

$$\mathcal{L}_{\text{pseudo}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}\mathbb{I}\Big(\hat{\mathbf{Y}}_i = c\Big)\log\left((\mathbf{Q}_{\text{prob}})_{i,c}\right), \tag{23}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

On top of the pseudo-label guidance, we further introduce a distribution regularization term to improve the structural rationality of clustering assignments and to avoid degenerate solutions:

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{conf}} + \mathcal{L}_{\text{bal}} + \mathcal{L}_{\text{smooth}}. \tag{24}$$

Specifically, the confidence loss $\mathcal{L}_{\text{conf}}$ minimizes the node-wise prediction entropy so that each superpixel tends to produce a more definite cluster assignment:

$$\mathcal{L}_{\text{conf}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}(\mathbf{Q}_{\text{prob}})_{i,c}\log\left((\mathbf{Q}_{\text{prob}})_{i,c}\right). \tag{25}$$

The class balance loss $\mathcal{L}_{\text{bal}}$ constrains the global clustering distribution to avoid collapse into a few dominant clusters:

$$\mathcal{L}_{\text{bal}} = \sum_{c=1}^{C}\bar{q}_c\log(\bar{q}_c), \tag{26}$$

where $\bar{q}_c = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{Q}_{\text{prob}})_{i,c}$ is the global assignment frequency of the c-th cluster.

In addition, we impose a spatial smoothness loss to exploit the local correlation between adjacent regions. Let $\mathbf{A}$ denote the adjacency matrix used for assignment regularization at the current training stage. This loss penalizes inconsistent clustering distributions between connected nodes and is formulated as

$$\mathcal{L}_{\text{smooth}} = \frac{1}{\sum_{i,j}\mathbf{A}_{i,j}}\sum_{i,j}\mathbf{A}_{i,j}\left\|(\mathbf{Q}_{\text{prob}})_i - (\mathbf{Q}_{\text{prob}})_j\right\|_2^2. \tag{27}$$

The predicted clustering distribution $\mathbf{Q}_{\text{prob}}$ plays two roles in the proposed framework. It is directly constrained by the clustering-related losses, and is also used to construct the semantic consistency term $\mathbf{S}_{\text{q}}$ for graph evolution. In this way, the current clustering semantics are incorporated as complementary guidance for topology refinement, rather than being used as the sole criterion for graph updating.

To encourage graph topology evolution to be consistent with interpretable multi-frequency guidance, we introduce an edge evolution loss to align the learnable prediction $\mathbf{W}_{\text{pre}}$ with the empirically guided matrix $\mathbf{W}_{\text{emp}}$. Let

$$\Omega = \{(i,j) \mid \mathbf{M}_{i,j} = 1\} \tag{28}$$

denote the set of initially connected local edges. The edge evolution loss is defined as

$$\mathcal{L}_{\text{edge}} = \frac{1}{|\Omega|}\sum_{(i,j)\in\Omega}\left(\mathbf{W}_{\text{pre},i,j} - \mathbf{W}_{\text{emp},i,j}\right)^2. \tag{29}$$

Here, $\mathbf{W}_{\text{pre}}$ denotes the learnable edge prediction derived from the current fused representation, while $\mathbf{W}_{\text{emp}}$ provides interpretable structural guidance derived from low-frequency affinity, high-frequency repulsion, and clustering distribution consistency. By minimizing $\mathcal{L}_{\text{edge}}$, the learned representation and graph evolution processes are explicitly coupled through shared multi-frequency evidence. Consequently, graph refinement is guided by both data-driven connectivity estimation and interpretable structural cues, encouraging the topology to evolve toward a more reliable structure during training.

E. Computational Complexity Analysis

The dominant computational cost of the proposed method comes from the collaborative multi-frequency representation module and the frequency-guided graph evolution module. Specifically, the Chebyshev graph filtering process requires repeated graph-operator multiplications, resulting in a complexity of approximately $O(KN^2d)$ under the dense graph formulation, where N denotes the number of superpixel nodes, d is the feature dimension, and K is the Chebyshev polynomial order. In addition, the graph evolution module performs pairwise node-relation modeling to predict learnable edge weights and construct empirical guidance, which introduces an additional complexity of approximately $O(N^2d)$. Therefore, the per-epoch complexity is approximately $O((K+1)N^2d)$, and the overall training complexity over T epochs is $O(T(K+1)N^2d)$. Since the graph topology is updated every T_u epochs, the reconstruction of graph filtering operators only introduces an amortized cost and does not change the leading-order complexity. Moreover, the model operates on a superpixel-level graph, where N is much smaller than the number of original pixels, making the framework computationally feasible in practice.

IV. EXPERIMENTS

Extensive experiments are conducted to evaluate the effectiveness of the proposed method for hyperspectral image clustering. We first introduce the datasets and experimental settings, including the compared methods, evaluation metrics,

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and hyperparameter configurations. Then, quantitative comparisons with representative state-of-the-art methods are reported, followed by ablation studies, hyperparameter analysis, and feature visualization. These evaluations are used to verify the proposed framework from the perspectives of clustering accuracy, component effectiveness, parameter robustness, and representation discriminability. Our code is available at https://github.com/Everything1025/SAMFGE.

A. Datasets and Experimental Setting

We conduct experiments on four widely used hyperspectral image datasets, including Pavia University (PU), Indian Pines (IP), WHU_Hi_HongHu, and WHU_Hi_LongKou. These datasets cover both urban and agricultural scenes with different spatial resolutions, spectral dimensions, scene complexities, and numbers of ground-truth land-cover categories, thereby providing a comprehensive evaluation of the proposed method. Specifically, PU and WHU_Hi_LongKou contain 9 land-cover categories, IP contains 16 land-cover categories, and WHU_Hi_HongHu contains 22 land-cover categories. The detailed information of these datasets is described as follows.

1) Pavia University (PU): The Pavia University dataset was acquired by the Reflectance Optical System Imaging Spectrometer (ROSIS) over Pavia, Northern Italy. It is a representative urban scene with a spatial size of 610×340 pixels. After removing 12 noisy bands, 103 spectral bands are retained, and the spatial resolution is 1.3 m.

2) Indian Pines (IP): The Indian Pines dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over northwestern Indiana, USA. It is a representative agricultural scene with a spatial size of 145×145 pixels. After removing water-absorption and noisy bands, 200 spectral bands are retained, and the spatial resolution is approximately 20 m.

3) WHU_Hi_HongHu: The WHU_Hi_HongHu dataset was collected by a DJI Matrice 600 Pro UAV equipped with a Headwall Nano-Hyperspec sensor in Honghu City, Hubei Province, China. It corresponds to a complex agricultural scene with diverse land-cover categories and intricate spatial structures. The data size is 940×475 pixels, with 270 spectral bands and a spatial resolution of about 0.043 m.

4) WHU_Hi_LongKou: The WHU_Hi_LongKou dataset was collected by a DJI Matrice 600 Pro UAV equipped with a Headwall Nano-Hyperspec sensor in Longkou Town, Hubei Province, China. It represents a relatively simple agricultural scene with a spatial size of 550×400 pixels, 270 spectral bands, and a spatial resolution of 0.463 m.

For comparison, we select nine recent representative hyperspectral image clustering methods, including NCSC [47], DSCRLE [21], DSGCC [48], SDST [49], SSGCC [50], S2GCL [51], LFGD [52], SLCGC [35], and SSGCO-EGAEL [39]. SSGCO-EGAEL is abbreviated as SSGCO in the tables and figures. These methods cover representative paradigms, including deep subspace clustering, graph-based deep clustering, self-supervised graph clustering, contrastive graph clustering, graph diffusion-based clustering, and graph structure refinement, providing a comprehensive comparison basis.

TABLE II
HYPERPARAMETER SETTINGS ON DIFFERENT DATASETS

Dataset	Training epochs T	Superpixel scale S	Chebyshev order K	Retained principal components d	Learning rate η
PU	400	600	4	10	5×10^{-4}
IP	400	90	5	55	1×10^{-3}
HongHu	400	650	4	16	5×10^{-4}
LongKou	400	600	4	20	1×10^{-3}

For the proposed method, five main hyperparameters are considered, namely the number of training epochs T, the superpixel scale parameter S, Chebyshev polynomial order K, which determines the order of the Chebyshev graph filter, the retained PCA dimension d and the learning rate η. Here, S controls the granularity of SLIC-based superpixel segmentation. For an HSI with spatial size $H \times W$, the initial number of SLIC segments is set as $n_{\text{segments}} = \lfloor HW/S \rfloor$, so that S approximately represents the expected average number of pixels within each superpixel. The used hyperparameter settings on different datasets are summarized in Table II, and the effects of the key parameters K, S and d are further analyzed in Section IV-D.

Following common practice in hyperspectral clustering, five widely used evaluation metrics are adopted, including accuracy (ACC), kappa coefficient (Kappa), normalized mutual information (NMI), adjusted Rand index (ARI), and Purity. Among them, ACC, NMI, and Purity range from 0 to 1, while Kappa and ARI range from -1 to 1. For clarity, all results are reported in percentage form, where higher values indicate better clustering performance. In addition, the Hungarian algorithm [53] is employed to align the predicted labels with the ground-truth labels before metric calculation. All compared methods are reproduced based on their official implementations. To ensure a fair and rigorous comparison, all methods are evaluated under a unified protocol, including the same input data, background-pixel exclusion rule, label alignment strategy, and metric calculation procedure. The predicted labels are consistently aligned with the ground-truth labels before evaluation.

B. Overall Clustering Results and Analysis

Table III reports the quantitative clustering results of the proposed method and nine representative baselines on the PU, IP, HongHu, and LongKou datasets. For each metric, the best result is highlighted in bold and the second-best result is underlined. Overall, the proposed method achieves the best performance on the majority of metric entries across the four datasets and remains competitive in the remaining cases, indicating its effectiveness under different hyperspectral scene characteristics.

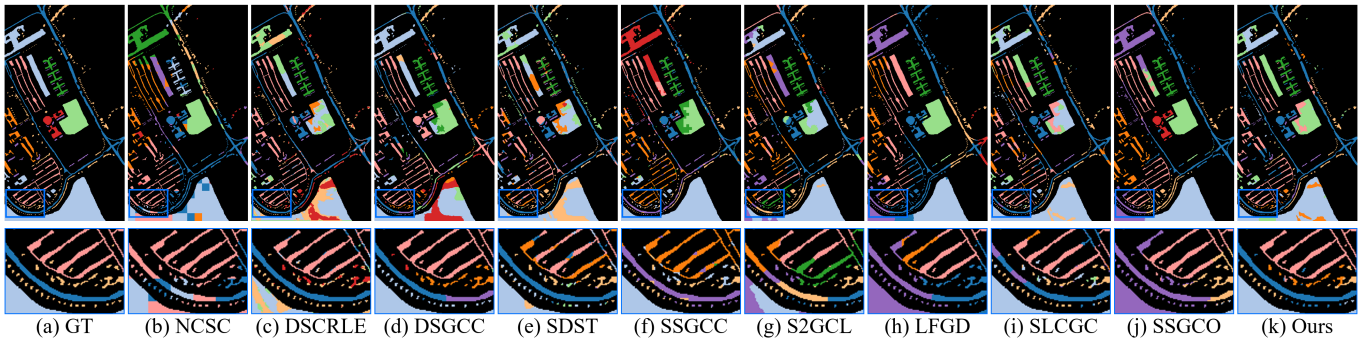
The performance improvement of the proposed method can be interpreted from the perspective of its design motivation. Many graph-based clustering methods mainly rely on smooth propagation, which enhances intra-region consistency but may weaken boundary-sensitive and locally heterogeneous structures. In contrast, the proposed method jointly models low-

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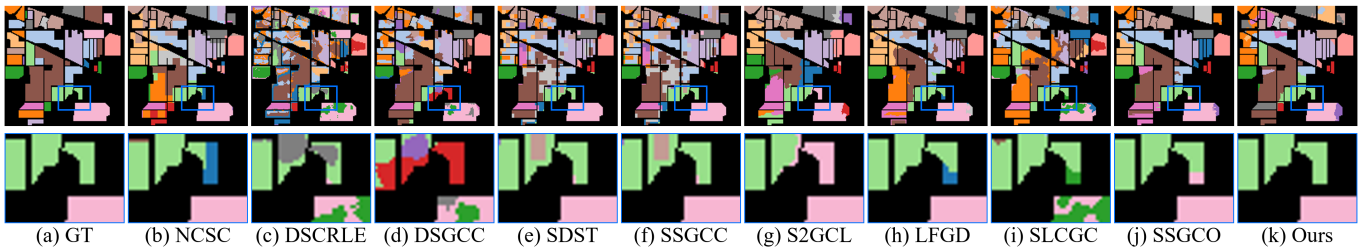
TABLE III
CLUSTERING PERFORMANCE ON FOUR HYPERSPECTRAL DATASETS

Dataset	Metric (%)	NCSC	DSCRLE	DSGCC	SDST	SSGCC	S2GCL	LFGD	SLCGC	SSGCO	Ours
PU	ACC	47.99	58.84	64.56	58.19	64.27	62.05	60.50	67.86	70.14	**74.25**
	Kappa	38.26	48.34	55.30	46.38	55.12	51.16	53.16	57.90	64.04	**67.12**
	NMI	42.40	58.58	57.38	56.15	59.49	52.04	56.69	60.19	65.93	**67.38**
	ARI	32.43	38.46	53.21	40.95	53.31	48.86	42.71	51.33	51.97	**60.54**
	Purity	63.34	71.35	74.56	69.76	75.86	71.13	76.27	72.95	**79.12**	77.16
IP	ACC	52.37	46.52	56.51	54.22	51.72	57.95	55.19	60.71	64.72	**66.36**
	Kappa	48.20	40.60	50.97	47.49	46.70	53.34	50.51	54.91	60.25	**62.06**
	NMI	56.85	47.84	57.63	51.68	53.28	61.71	59.05	62.61	65.13	**66.06**
	ARI	33.97	29.44	39.46	34.63	32.84	42.59	36.20	41.13	49.65	**50.28**
	Purity	63.79	57.59	64.35	57.62	61.77	64.88	64.37	63.51	72.21	**73.97**
HongHu	ACC	39.63	40.20	62.35	50.26	51.34	56.17	58.51	58.30	54.78	**79.70**
	Kappa	35.93	34.36	54.85	43.16	45.52	50.35	53.71	52.92	49.47	**74.53**
	NMI	59.18	52.64	66.30	58.57	60.10	67.61	70.41	68.48	68.01	**75.03**
	ARI	21.79	27.17	65.70	46.18	34.69	47.79	45.53	46.11	46.82	**86.96**
	Purity	77.51	74.26	77.35	75.16	72.85	80.74	81.61	81.26	82.07	**82.59**
LongKou	ACC	45.18	68.32	71.61	72.58	57.10	66.81	62.98	72.56	62.20	**78.17**
	Kappa	36.62	60.85	64.87	66.10	48.98	60.00	54.54	66.61	54.78	**72.27**
	NMI	50.78	70.12	72.87	71.53	65.35	63.83	63.33	72.86	63.45	**77.90**
	ARI	55.98	69.79	73.70	66.72	50.35	58.65	52.96	71.32	50.77	**74.83**
	Purity	74.76	85.98	86.97	86.75	84.83	80.07	83.72	88.46	80.18	**90.45**



(a) GT (b) NCSC (c) DSCRLE (d) DSGCC (e) SDST (f) SSGCC (g) S2GCL (h) LFGD (i) SLCGC (j) SSGCO (k) Ours

Fig. 3. Clustering maps of the PU dataset. (a) GT, (b) NCSC, (c) DSCRLE, (d) DSGCC, (e) SDST, (f) SSGCC, (g) S2GCL, (h) LFGD, (i) SLCGC, (j) SSGCO, and (k) Ours.



(a) GT (b) NCSC (c) DSCRLE (d) DSGCC (e) SDST (f) SSGCC (g) S2GCL (h) LFGD (i) SLCGC (j) SSGCO (k) Ours

Fig. 4. Clustering maps of the IP dataset. (a) GT, (b) NCSC, (c) DSCRLE, (d) DSGCC, (e) SDST, (f) SSGCC, (g) S2GCL, (h) LFGD, (i) SLCGC, (j) SSGCO, and (k) Ours.

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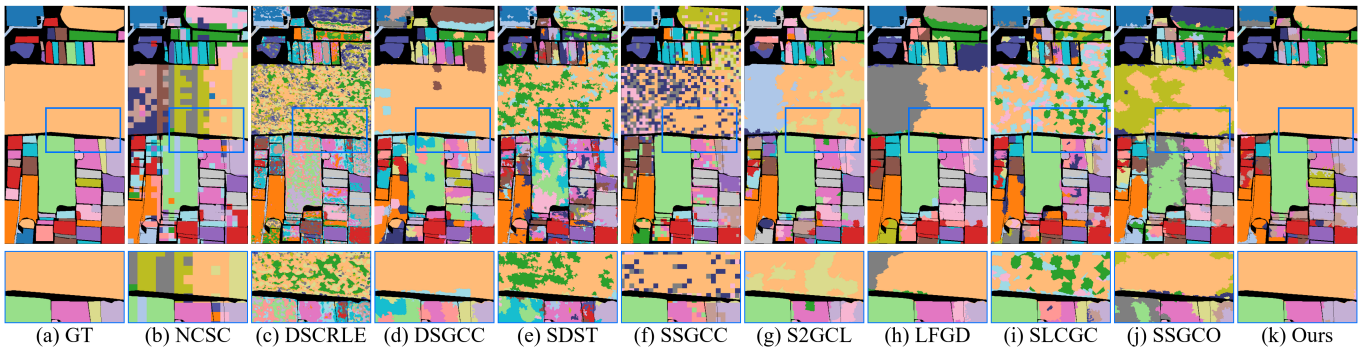


Fig. 5. Clustering maps of the HongHu dataset. (a) GT, (b) NCSC, (c) DSCRLE, (d) DSGCC, (e) SDST, (f) SSGCC, (g) S2GCL, (h) LFGD, (i) SLCGC, (j) SSGCO, and (k) Ours.

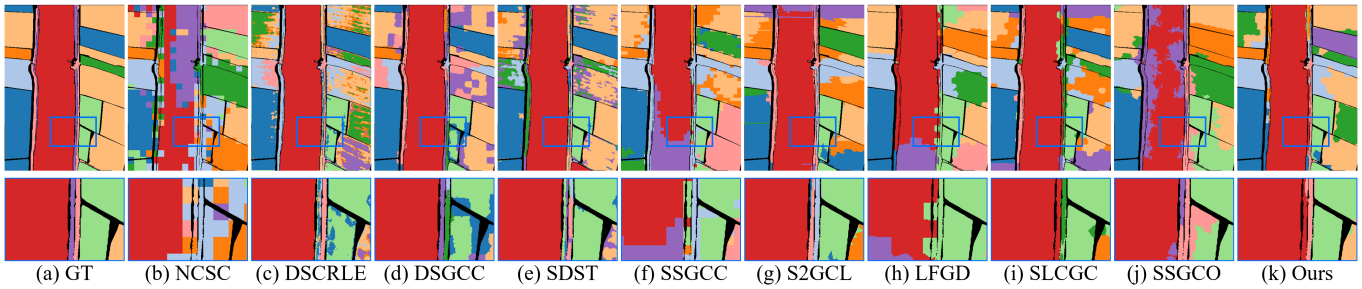


Fig. 6. Clustering maps of the LongKou dataset. (a) GT, (b) NCSC, (c) DSCRLE, (d) DSGCC, (e) SDST, (f) SSGCC, (g) S2GCL, (h) LFGD, (i) SLCGC, (j) SSGCO, and (k) Ours.

frequency smooth consistency and high-frequency heterogeneous responses within a unified multi-frequency representation framework. This design helps enhance regional coherence while retaining boundary-sensitive and locally heterogeneous information, thereby producing more discriminative node representations for clustering.

Moreover, multi-frequency cues are not only used for node representation learning, but are also exploited as structural evidence for graph evolution. By using low-frequency affinity and high-frequency discrepancy to regularize edge updating, the proposed method can reduce unreliable cross-boundary propagation caused by erroneous initial connections. Therefore, its advantage comes from the coordinated optimization of frequency-aware representation learning and graph topology refinement.

Figs. 3–6 present the clustering maps of different methods on the four datasets. Several compared methods still suffer from fragmented predictions in homogeneous regions, blurred boundaries, or insufficient preservation of local structures. In contrast, the proposed method produces clustering maps that are more consistent with the ground-truth spatial distributions. It yields more spatially coherent regions with fewer scattered misclassifications, while better preserving boundary information and local structural details. These qualitative observations are consistent with the quantitative results in Table III and further support the effectiveness of the proposed multi-frequency collaborative framework.

C. Ablation Studies

To evaluate the effectiveness of the proposed components, we conduct ablation studies by removing or disabling each key module from the complete framework. Specifically, the evaluated components include the low-frequency homogeneity modeling (LFM), the high-frequency heterogeneity modeling (HFM), the node- and channel-aware frequency coordination (NCAFC), and the frequency-guided graph evolution (FGGE). The corresponding variants are defined as follows: *w/o LFM* removes the low-frequency homogeneity modeling while the remaining modules are retained, *w/o HFM* removes the high-frequency heterogeneity modeling to evaluate the contribution of high-frequency structural responses. *w/o NCAFC* disables the node- and channel-aware frequency coordination module and directly fuses the multi-frequency representations without adaptive frequency selection, and *w/o FGGE* keeps the collaborative multi-frequency representation module but disables the frequency-guided graph evolution module. The complete model is denoted as *Full*. For a concise and consistent evaluation, three representative metrics, namely ACC, Kappa, and ARI, are reported, since they respectively reflect clustering accuracy, chance-corrected agreement, and pairwise partition consistency.

The ablation results are summarized in Table IV. It can be observed that the complete model obtains the best results across all reported metrics on the four datasets, indicating the effectiveness of jointly incorporating the proposed components. More importantly, the performance variations of different ablation variants provide further insight into the role of each module within the overall framework.

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TABLE IV
ABLATION STUDY OF KEY COMPONENTS ON FOUR HSI DATASETS

Module				PU			IP			HongHu			LongKou		
LFM	HFM	NCAFC	FGGE	ACC	Kappa	ARI	ACC	Kappa	ARI	ACC	Kappa	ARI	ACC	Kappa	ARI
✗	✓	✓	✓	69.20	60.12	49.45	50.12	46.03	31.36	73.14	66.74	80.84	41.58	30.27	22.88
✓	✗	✓	✓	66.17	55.55	51.35	45.21	39.21	29.67	70.91	64.38	70.39	76.45	70.44	71.51
✓	✓	✗	✓	68.12	57.53	53.31	63.73	58.82	44.34	77.02	71.61	80.92	76.97	70.83	73.16
✓	✓	✓	✗	72.38	64.45	57.62	63.66	59.11	47.23	76.82	71.24	85.21	76.02	66.34	59.89
✓	✓	✓	✓	**74.25**	**67.12**	**60.54**	**66.36**	**62.06**	**50.28**	**79.70**	**74.53**	**86.96**	**78.17**	**72.27**	**74.83**

✓ indicates that the corresponding component is included, while ✗ indicates that it is removed.

First, removing LFM leads to clear performance degradation across the four datasets, with a particularly pronounced drop on LongKou. This indicates that low-frequency homogeneity modeling is essential for capturing intra-region consistency and stabilizing homogeneous-region representations. Without this component, the model loses an important basis for preserving regional coherence, which weakens the discriminability of the learned clustering representations.

Second, removing HFM also causes noticeable performance drops on all datasets. This suggests that high-frequency structural cues provide important complementary information for boundaries, transition regions, and locally heterogeneous structures. In particular, the degradation on IP and HongHu shows that relying only on smooth consistency is insufficient for preserving the structural details that are important for accurate clustering, further supporting the necessity of jointly modeling low-frequency homogeneity and high-frequency heterogeneity in hyperspectral graph clustering.

Third, removing NCAFC leads to consistent performance degradation on all datasets. Although this variant retains both frequency branches, it disables adaptive coordination at the node and channel levels. This result indicates that simply extracting multi-frequency representations is not sufficient. Properly assigning frequency preferences according to local structural properties and discriminative channels is important for fully exploiting the complementarity between low and high-frequency cues.

Fourth, removing FGGE also results in consistent performance degradation, indicating the benefit of frequency-guided graph evolution. Since this variant retains the collaborative multi-frequency representation module, the performance drop mainly reflects the contribution of using learned frequency cues to refine graph topology. This suggests that FGGE improves graph propagation reliability by exploiting low-frequency affinity and high-frequency discrepancy for edge updating.

Overall, the ablation results indicate that the LFM and HFM play complementary roles in multi-frequency representation learning. NCAFC further improves their integration by adaptively balancing frequency preferences according to node-wise structural properties and channel-wise discriminative spectral-spatial responses. Meanwhile, FGGE provides additional support for graph structure optimization by exploiting the learned multi-frequency cues, thereby further enhancing clustering performance.

D. Parameter Sensitivity Analysis

The proposed method involves five main hyperparameters, namely the number of training iterations T, the superpixel scale parameter S, Chebyshev polynomial order K, the retained PCA dimension d and the learning rate η. Among them, K directly controls the receptive field of Chebyshev graph filtering and is closely related to the proposed multi-frequency modeling mechanism. Therefore, we analyze its influence by varying K from 2 to 6 while keeping the other hyperparameters fixed.

As shown in Fig. 7, the proposed method achieves the best overall performance at $K = 4$ on PU, HongHu, and LongKou, and at $K = 5$ on IP. In general, the performance improves as K increases from a small value to a moderate range, but may decline when K becomes excessively large. This trend is consistent with the role of K in Chebyshev graph filtering. Specifically, K denotes the Chebyshev polynomial order and controls the effective receptive field of localized graph propagation. A small K restricts the propagation range and limits contextual information aggregation, whereas a moderate K helps capture useful multi-hop structural information. However, an excessively large K may introduce redundant propagation and weaken local structural distinctions, thereby reducing the discriminability of the learned representations.

We further evaluate the sensitivity of the superpixel scale parameter S, which controls the granularity of SLIC-based superpixel segmentation and affects the construction of region-level graph nodes. Since the four datasets have different spatial sizes, we use the normalized relative granularity $r_s = \frac{S}{H \times W} \times 100\%$ for visualization. As shown in Fig. 8, the proposed method achieves relatively stable performance when S is set within a moderate range. An excessively small S may produce fragmented superpixels, while an excessively large S may merge different land-cover structures into coarse regions. Both cases can weaken the reliability of graph representations and affect clustering performance.

We additionally investigate the influence of the PCA dimension d. As shown in Fig. 9, although the preferred value of d varies across datasets, the clustering performance generally changes smoothly with d, indicating that our method remains relatively stable under moderate variations in this parameter. In general, retaining too few principal components

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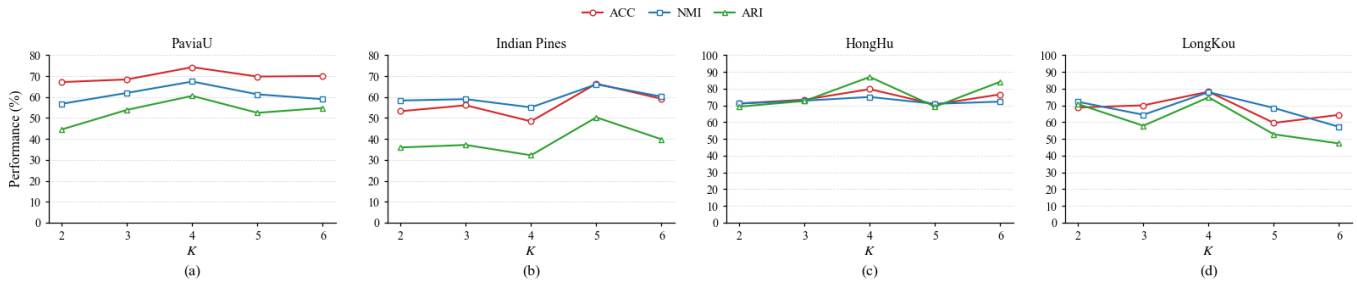


Fig. 7. Sensitivity analysis of parameter K on different datasets.

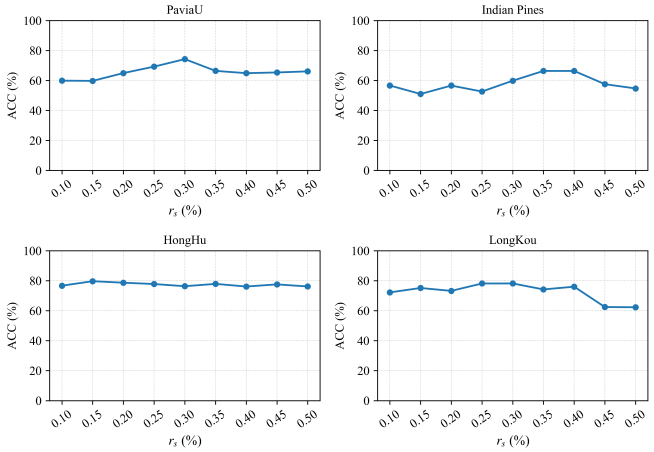


Fig. 8. Sensitivity analysis of parameter S on different datasets, where the relative granularity is normalized as $r_s = \frac{S}{H \times W} \times 100\%$.

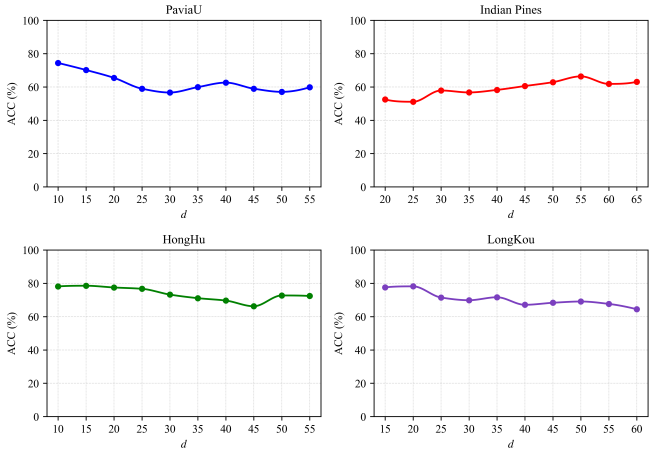


Fig. 9. Sensitivity analysis of parameter d on different datasets.

may discard spectral variations useful for distinguishing neighboring regions, whereas retaining more components preserves additional information but may also introduce noise or less discriminative variations.

Overall, moderate settings of K and S, together with an appropriate PCA dimension d are more suitable for the proposed framework, as they provide a better balance between multi-frequency information aggregation, superpixel-level structural reliability, and the preservation of discriminative local cues.

E. Representation Visualization

To intuitively evaluate the representation learning capability of the proposed method, we employ t-distributed stochastic neighbor embedding (t-SNE) to visualize the original pixel features and the learned representations on all four datasets. The learned superpixel-level node representations are assigned to the corresponding pixels for visualization. Fig. 10 compares the distributions of the original input features with those of the mapped learned representations.

In the original feature space, some categories show local aggregation, but several classes still exhibit overlapping or elongated distributions, indicating limited separability among land-cover categories. After representation learning, the visualized representations generally become visually more compact within classes and more separated across classes. Several categories form clearer cluster structures, and the boundaries between different clusters become more distinguishable in the embedding space. The learned representations achieve higher silhouette scores than the original pixel features on all datasets.

TABLE V
RUNNING TIME (SECONDS) COMPARISON ON FOUR DATASETS.

Method	PU	IP	HongHu	LongKou	Mean
NCSC	2140.7	1212.4	4381.3	2536.6	2567.8
DSCRLE	3639.8	1429.1	7678.8	1689.6	3609.3
DSGCC	13.7	6.9	46.7	24.3	22.9
SDST	1264.6	502.5	9083.1	2532.9	3345.8
SSGCC	213.9	52.4	407.6	191.1	216.3
S2GCL	15.5	4.9	45.5	28.4	23.6
LFGD	63.9	16.5	136.2	60.3	69.2
SLCGC	80.8	10.7	187.9	49.3	82.2
SSGCO	135.8	90.7	258.3	101.4	146.6
Ours	22.3	8.2	128.6	33.2	48.1

These observations are consistent with the design motivation of the proposed framework. By coordinating low-frequency homogeneous consistency and high-frequency structural cues, the proposed method helps preserve intra-class coherence while enhancing inter-class separability. Therefore, the t-SNE visualization provides intuitive evidence that the proposed framework can learn more discriminative representations for hyperspectral image clustering.

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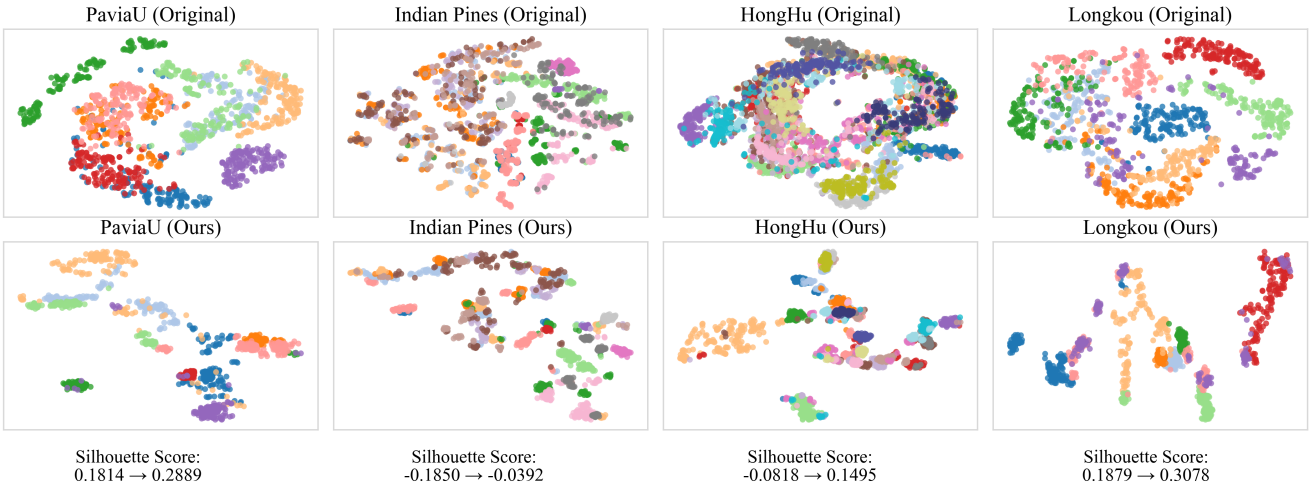


Fig. 10. t-SNE visualization of original pixel features and mapped learned representations on different datasets.

F. Running Time Analysis

To further evaluate computational efficiency, we report the runtime of all methods on the four datasets in Table V under the same experimental environment and timing protocol. Although DSGCC and S2GCL are faster, our method provides a favorable balance between computational efficiency and clustering performance. Its efficiency mainly benefits from superpixel-level graph representation, which reduces the graph scale, and Chebyshev polynomial approximation, which avoids explicit eigendecomposition. Overall, SAMFGE achieves strong clustering performance with moderate computational cost.

V. CONCLUSION

This paper proposed a spatially adaptive multi-frequency graph representation and evolution framework for hyperspectral image clustering. The motivation is that hyperspectral scenes contain heterogeneous spatial structures with different graph-frequency demands, where homogeneous regions require low-frequency consistency while boundaries, mixed regions, and fine-grained structures benefit from high-frequency discrimination. To address this issue, the proposed method learns complementary low-frequency and high-frequency representations on a region-level graph, and adaptively coordinates them at both the node and channel levels. In addition, the learned multi-frequency cues are further exploited as interpretable structural evidence to guide graph evolution, improving the reliability of graph propagation. Experimental results on multiple public hyperspectral datasets demonstrate that the proposed framework can effectively improve clustering performance. Despite the encouraging results, there remains room for further improving the robustness of high-frequency modeling. Some noise-induced local variations may still exhibit frequency characteristics similar to meaningful structural changes. Future work will investigate a reliability-aware high-frequency modeling strategy that combines cross-scale response consistency, spatial-neighborhood coherence, and spectral contrast as complementary evidence for reweighting high-frequency responses, thereby preserving informative boundary variations while suppressing isolated or inconsistent responses that are more likely to arise from noise.

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